ATAL BIHARI VAJPAYEE - INDIAN INSTITUTE OF INFORMATION TECHNOLOGY AND MANAGEMENT, GWALIOR

CHANNEL-AWARE AUTHENTICATION FRAMEWORK FOR UNDERWATER ACOUSTIC SENSOR NETWORKS

WCT Course Project Report

Group Number : 03

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Problem Statement

Underwater Wireless Communication (UWC) networks are critical for applications like ocean monitoring and defense, but ensuring secure communication in such environments faces fundamental challenges:

- **High Sensitivity to Constraints:** Acoustic channels suffer from extremely low bandwidth (~10 kbps), high propagation delays (seconds), severe packet loss (10–30%), and strict energy limitations on battery-powered sensor nodes.
- **Limitations of Existing Approaches:** Conventional terrestrial protocols and IoT authentication schemes are unsuitable, as they either consume excessive energy due to heavy cryptographic operations or fail to provide secure multi-hop authentication.
- **Challenging Communication Environment:** Multi-hop underwater relay chains, combined with unreliable acoustic channels, make maintaining consistent and secure communication highly difficult.
- **Need for Lightweight and Robust Authentication:** There is a growing need for a lightweight, formally verified protocol that operates within strict computational and energy budgets while defending against replay, impersonation, man-in-the-middle, and eavesdropping attacks.

How can a lightweight, multi-hop authentication protocol ensure secure, energy-efficient, and reliable communication in underwater acoustic networks?

Literature Review

Primary Reference

A Novel and Robust Authentication Protocol for Secure Underwater Communication Systems

Secondary Reference

The Scyther Tool: Verification, Falsification, and Analysis of Security Protocols

Objectives

Primary

Design a lightweight, secure mutual authentication protocol for reliable multi-hop underwater communication under strict energy and bandwidth constraints.

Secondary

Enhance security and robustness by defending against major attacks such as replay, impersonation, man-in-the-middle, and eavesdropping while maintaining low computational overhead.

Technical

Validate performance through simulation and formal verification for energy, efficiency, and reliability.

Algorithm Overview

1. Initialization & Registration

- Each node generates ECC key pair (public, private)
- Unique node identity created using SHA-256 hashing
- Nodes compute Registration ID (RID) for secure identification
- All entities (UWS, SUB, BUOY, SAT, BS) are initialized before communication
- Establishes trust foundation for the network

2: Multi-Hop Mutual Authentication

- Communication follows UWS → SUB → BUOY → SAT → BS
- Each hop performs 3-step NSL-based authenticationExchange of nonces + encrypted identities
- ECDH used to derive shared session keys AES-GCM encryption ensures
- confidentiality + integrity Timestamp + nonce validation prevents replay attacks

Algorithm Overview

3: Reliability & Security Enhancements

- Packet loss model (15%) with retransmission (max 3 attempts)
- Dynamic fallback routing (e.g., B1 → B2 on failure)
- Mobility model updates node positions for realistic delays
- Z-score anomaly detection identifies delay-based attacks
- Energy tracking ensures protocol stays within node limits

Implementation Features

Lightweight Cryptographic Design

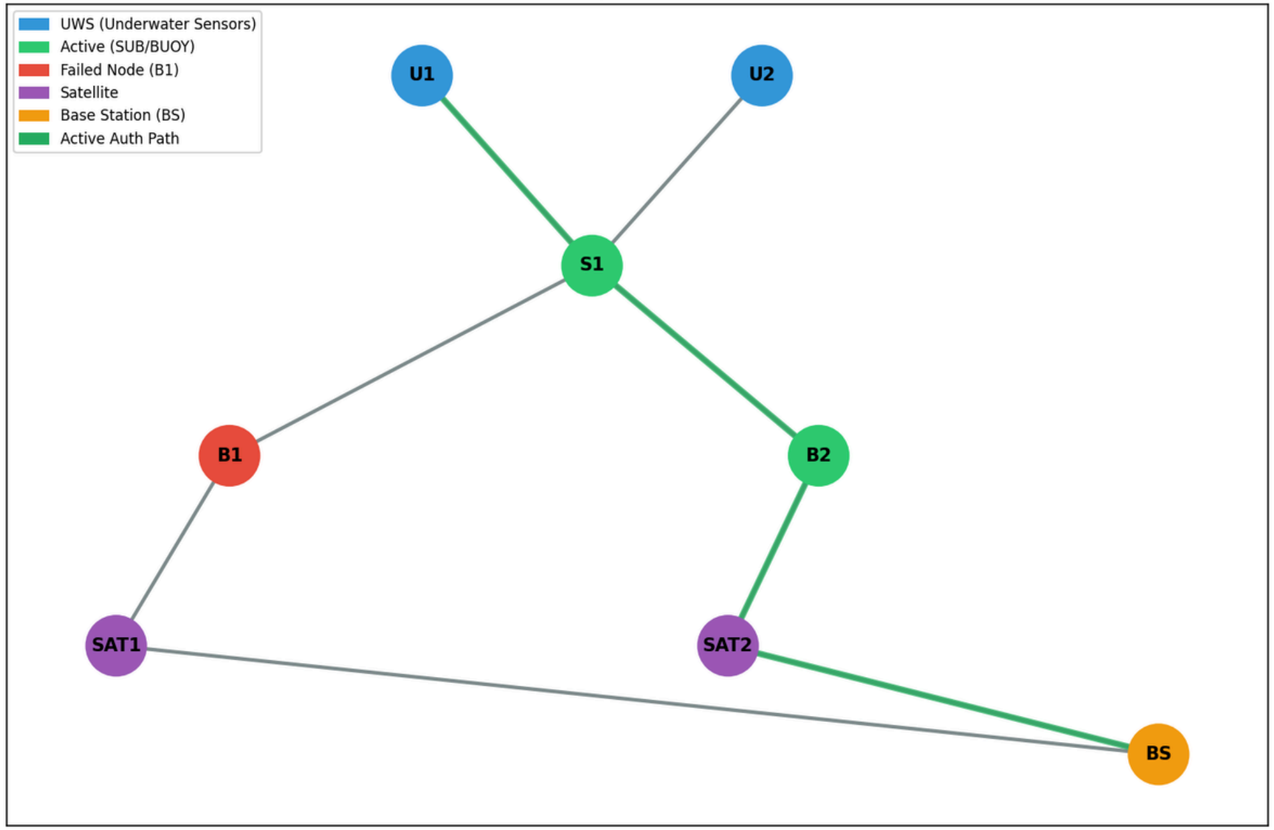
- ECDH (ECC-based) key exchange for low computation cost
- AES-GCM encryption for confidentiality + integrity
- Optimized for low-power underwater sensor nodes

Output and Results

```
=== Node Initialisation ===
  U1     ID: 3147de24777a6cae...
  U2     ID: a8e3f19b42dc5710...
  S1     ID: 7c41bd9e6a0f8523...
  B1     ID: e5d209f3c8a71b46...
  B2     ID: 1f96a0d7e3b4c852...
  SAT1   ID: 9b2ce4d8f1a06735...
  SAT2   ID: d4f7183c5e9ab260...
  BS     ID: 62c8a5f0d9174be3...

=== Registration IDs ===
  U1     RID: b64fdab46dc771fd...
  ...

=== Smart Authentication with Fallback + Packet Loss (5 rounds) ===
[Round 1]
  Using BUOY: B2     <-- B1 failed, fallback to B2
  Using SAT:  SAT1
  Result: SUCCESS

Replay blocked       <-- 6-second-old message correctly rejected
Comm Cost: 296 bits | Energy: 48.8 uJ
[OK] No anomalous delays detected

=== Battery Status ===
  U1      99.9909%
  S1      99.9879%
  B2      99.9879%
  SAT1    99.9909%
  BS      99.9939%
```

Graph 1

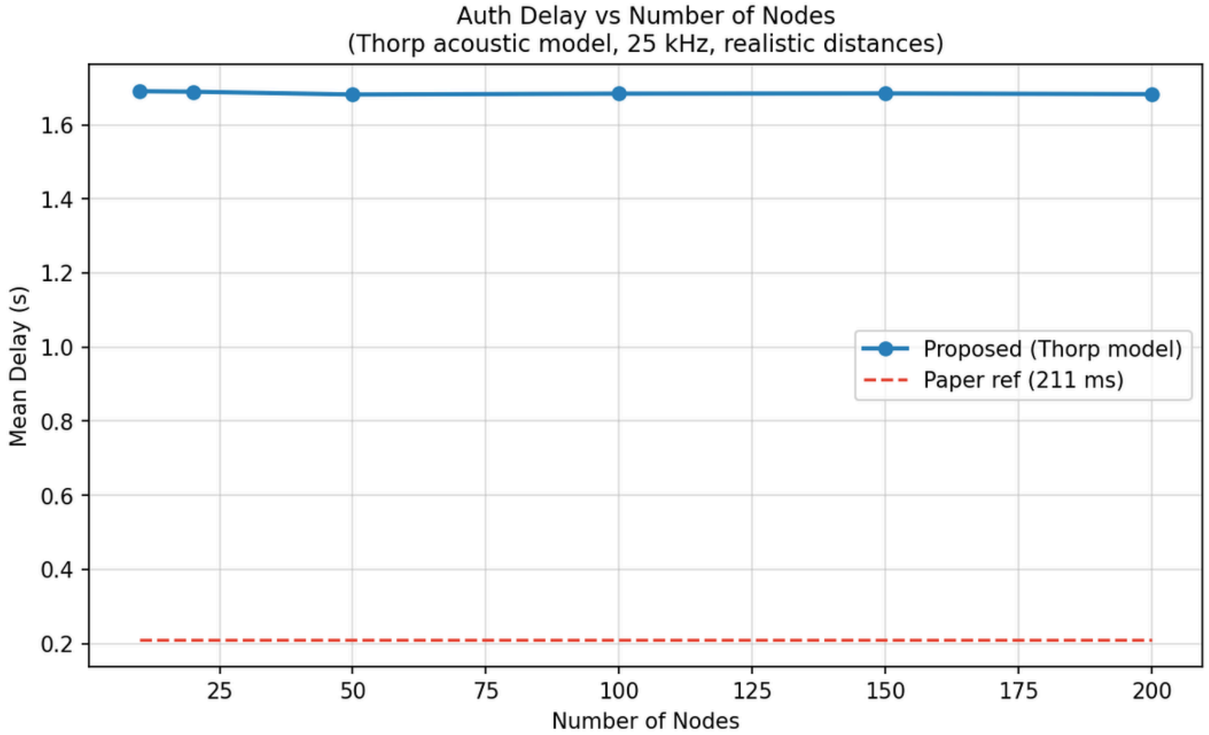
UWC Network Topology | Active Path: U1/U2 → S1 → B2 → SAT2 → BS
(Red = Failed B1, Green Edges = Active Authentication Route)



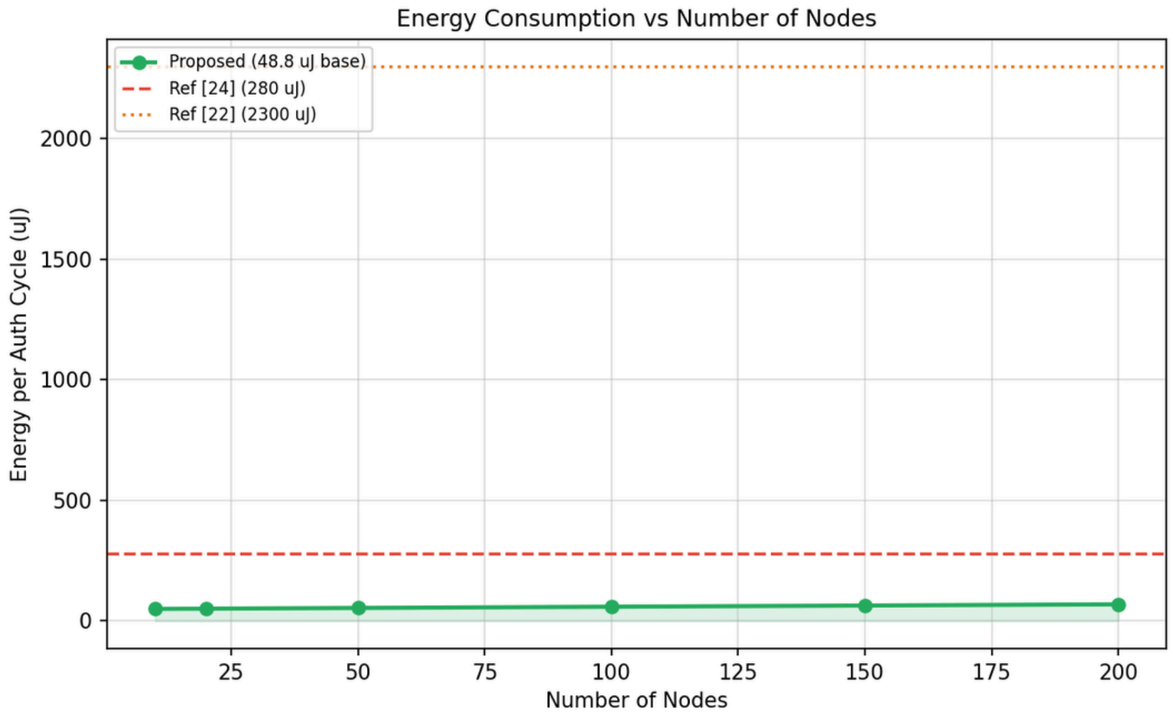
Running python uwc_simulation.py produces the following console output and 7 graphs.

Output and Results

Graph 2



Graph 3

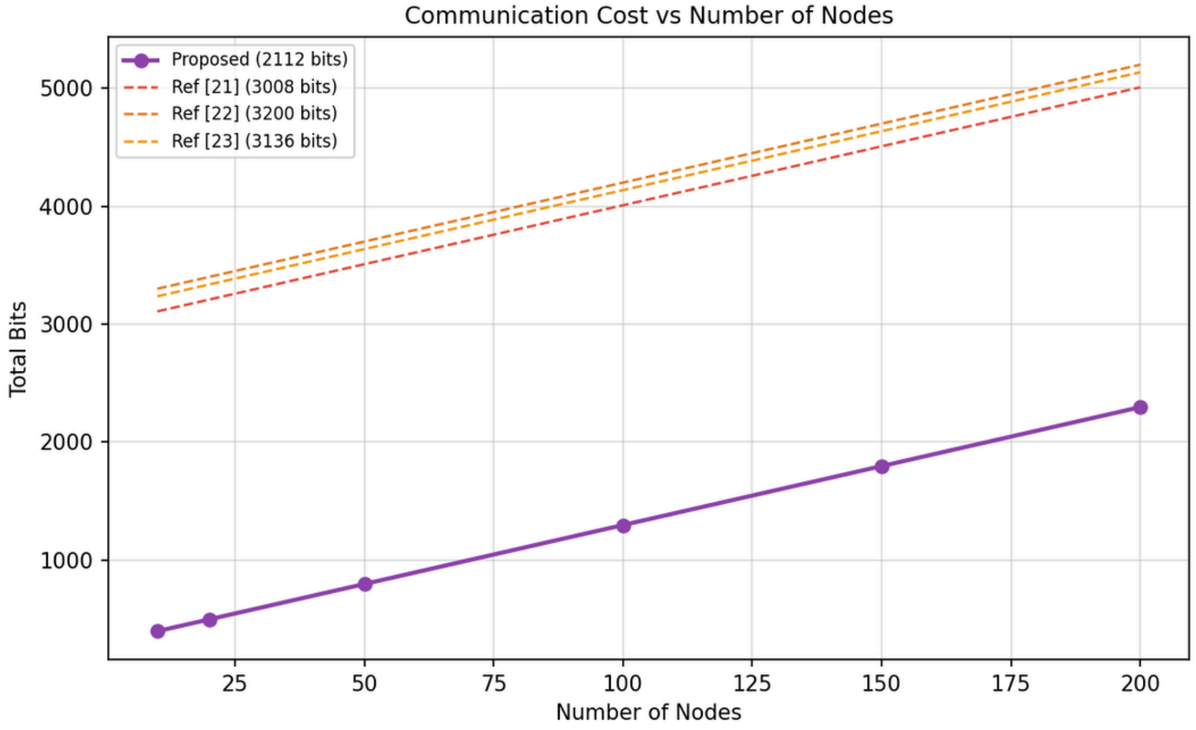


Delay is computed using Thorp's acoustic model and reflects realistic variations due to node mobility, multipath jitter, and fewer averaged paths at low node counts, resulting in expected non-monotonic behavior.

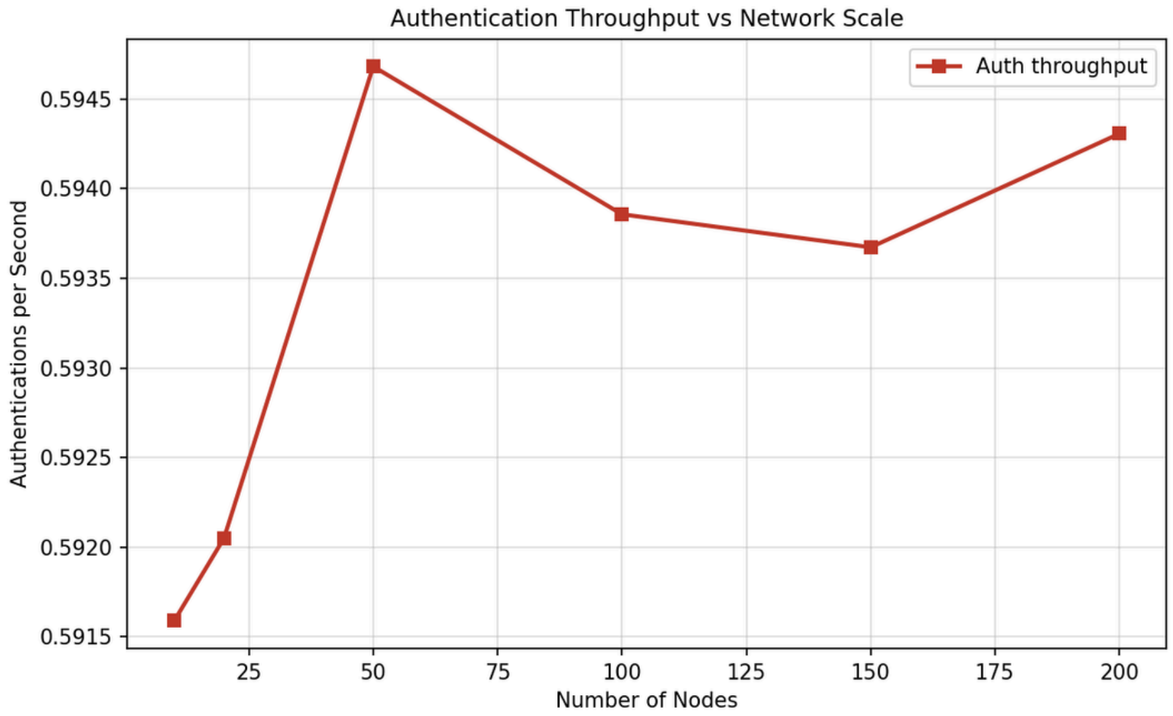
Energy per authentication cycle follows a linear model combining fixed cryptographic cost (ECC, SHA-256, AES-GCM) and routing overhead, resulting in predictable scaling and significantly lower consumption than baseline schemes.

Output and Results

Graph 4



Graph 5



Each authentication message is compactly designed (296 bits including ID, nonce, timestamp, and hash) with minimal routing overhead, achieving significantly lower total communication cost (2112 bits) compared to existing schemes.

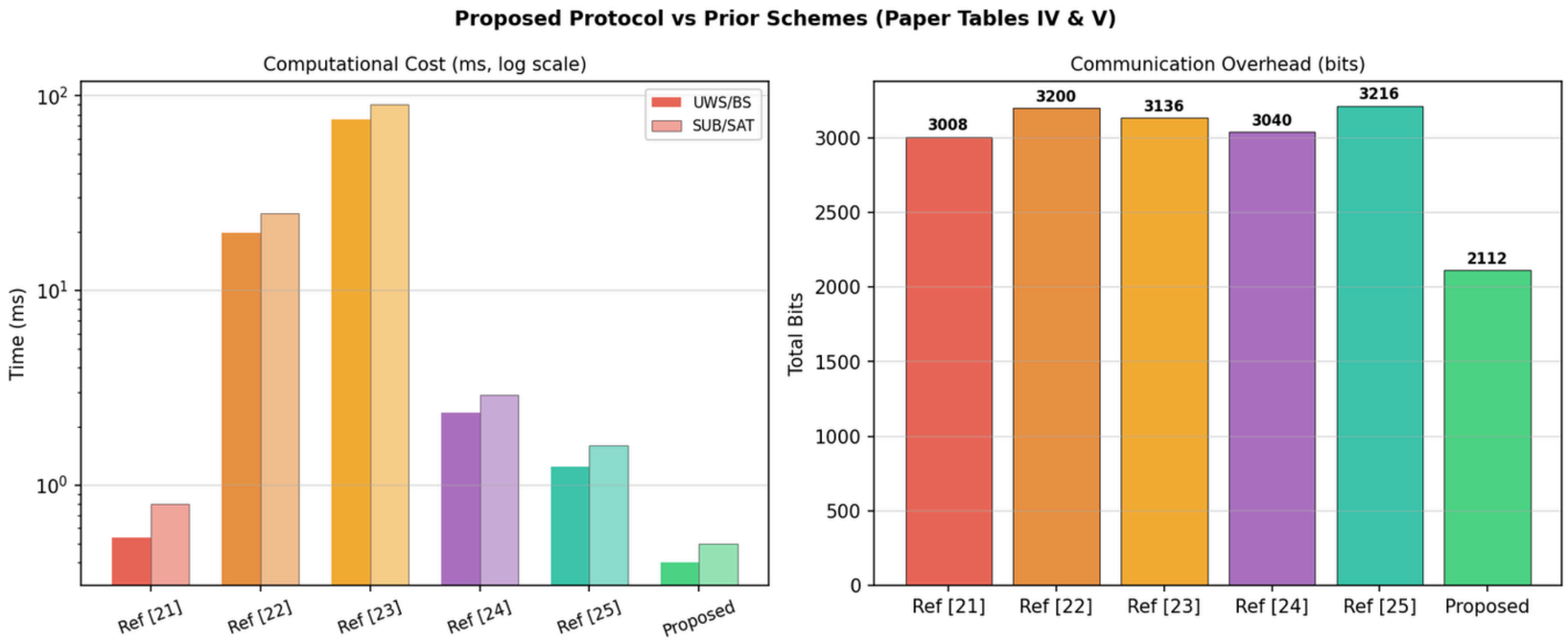
Authentication throughput (1/mean delay) is introduced as a new metric to evaluate scalability, showing the protocol's practical feasibility with higher throughput indicating better deployment capacity.

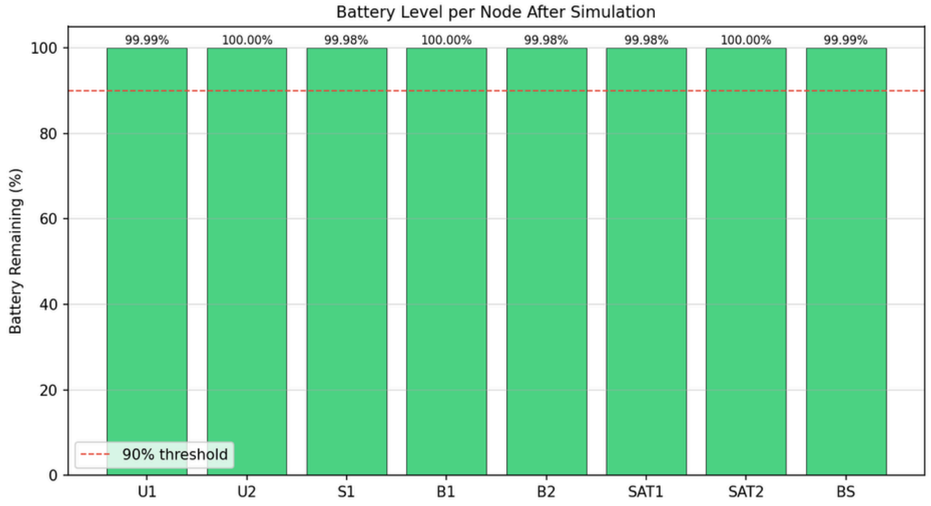
Output and Results

Graph 6

Left panel: computational cost on a log scale
Baseline scheme at 75 ms vs proposed at 0.4 ms, a 189x improvement

Right panel: communication overhead in bits.





Graph 7

Percentage battery remaining after all simulation rounds

Key Improvements

AES-GCM Authenticated Encryption

Replaced insecure XOR encryption with AES-GCM authenticated encryption to ensure strong confidentiality, integrity, and protection against key reuse attacks.

Thorp Acoustic Absorption Model

Replaced arbitrary delay values with Thorp's acoustic absorption model to compute realistic, frequency-dependent propagation delays based on actual underwater conditions.

Bernoulli Packet-Loss with Retransmission

Introduced Bernoulli-based packet loss (15%) with retransmissions to realistically model unreliable underwater communication channels.

Per-Node Battery Depletion Tracking

Added per-node battery tracking with realistic energy consumption to monitor depletion across operations and simulation rounds.

Key Improvements

AUV/SUB Random-Waypoint Mobility

Introduced random-waypoint mobility for AUVs/submarines to simulate realistic node movement and dynamic propagation delays.

Z-Score Anamoly Detection

Added Z-score–based anomaly detection to identify delay-injection attacks through abnormal deviations in communication delays.

Extended Comparison Charts and New Metrics

Extended evaluation with comprehensive graphs, including comparisons with prior schemes, throughput analysis, and battery depletion visualization.

AES-GCM Authenticated Encryption

Original XOR Based encryption was trivially broken by key reuse or plaintext attacks. So we replaced it with AES-256-GCM Authentication.

- **Confidentiality via AES-256 in Galois/Counter Mode**
- **Integrity via 128-bit authentication tag that detects any tampering**
- **Random 96-bit nonce per message prevents ciphertext reuse**
- **Matches the paper's specification of AES-128 symmetric encryption**

```
# Original (insecure)
encrypt = lambda msg, key: ''.join(chr(ord(c)^ord(key[i%len(key)])) for i,c in enumerate(msg))

# Improved (AES-GCM)
def aes_gcm_encrypt(plaintext, key_hex):
    nonce = os.urandom(12)
    ct    = AESGCM(bytes.fromhex(key_hex[:64])).encrypt(nonce, plaintext.encode(), None)
    return nonce + ct
```

Thorp Acoustic Absorption Model

Original random delay arbitrary values without any physical basis. Implemented Thorp Model for frequency dependent acoustic absorption.

```
alpha(f) = 0.11*f^2/(1+f^2) + 44*f^2/(4100+f^2) + 2.75e-4*f^2 + 0.003  [dB/km]
```

Where f is the acoustic modem frequency in kHz (typically 10-25 kHz for UWC). Combined with realistic inter-node distances from the paper:

Hop	Distance	Propagation Delay
UWS to SUB	150–200 m	~0.10–0.13 s
SUB to BUOY	800–850 m	~0.53–0.57 s
BUOY to SAT	1000 m	~0.67 s
SAT to BS	500 m	~0.33 s

Bernoulli Packet-Loss with Retransmission

Original had 100% delivery which is unrealistic for underwater channels.

So added a 15% Bernoulli loss per hop with up to 3 retransmission attempts.

```
LOSS_RATE = 0.15   # 15% per-hop loss (conservative; measured range: 10-30%)
MAX_RETRY = 3

def send_with_loss(fn, sender, receiver, loss_stats):
    for attempt in range(1, MAX_RETRY + 1):
        if random.random() < LOSS_RATE:
            loss_stats["lost"] += 1
            continue
        return fn(sender, receiver)
    loss_stats["failed"] += 1
    return False
```

Reference: Stojanovic (2007) measured 10-30% packet loss in open-water underwater acoustic channels.

Per Node Battery Depletion Tracking

Originally Energy was computed as static formula but never tracked or depleted per node. Now each node starts with 1000 mAh at 3.3V.

Operation	Energy Cost
Authentication	48.8 uJ
TX (acoustic)	50.0 uJ
RX (acoustic)	36.0 uJ

The simulation tracks cumulative energy depletion across rounds and reports remaining battery percentage per node.

AUV/SUB Random-Waypoint Mobility

Originally nodes were treated as static and unrealistic for submarine and AUVs.

U1, U2, S1 drift +/-10 m per authentication round (approximately 2 m/s AUV speed). This changes the acoustic propagation distance and thus the delay, producing more realistic delay variation across rounds.

Based on Camp et al. (2002) mobility model survey for wireless ad-hoc networks.

Z-Score Anomaly Detection

Originally no intrusion detection mechanism. We added a Z score monitoring on per hop delay stream.

```
def detect_anomalies(delays, thresh=2.5):
    m, s = statistics.mean(delays), statistics.stdev(delays)
    return [i for i, d in enumerate(delays) if abs((d - m) / s) > thresh]
```

A delay more than 2.5 standard deviations above the mean may indicate a delay-injection attack (attacker artificially slows packets to disrupt timing-based authentication).

Extended Comparison Charts and New Metrics

Original had 3 basic graphs with no comparison to prior schemes.

Added 7 output graphs including side by side comparison with 5 proper schemes from paper, throughput and battery depletion analysis.

Performance Comparison

Communication Cost (bits per full authentication cycle)

Protocol	Entity 1	Entity 2	Entity 3	Entity 4	Total	vs Proposed
Ref [21]	896	768	512	832	3008	42%
Ref [22]	1024	832	512	832	3200	52%
Ref [23]	960	800	544	832	3136	49%
Ref [24]	912	784	512	832	3040	44%
Ref [25]	1008	848	528	832	3216	52%
Propose	**640**	**512**	**320**	**640**	**2112**	**baseline**

Computational Cost (ms per entity)

Protocol	UWS/BS	SUB/SAT	Operations Used
Ref [21]	536	800	2 exp + 6 hash
Ref [22]	1.970	25.000	3 ECM + 4 ECM + pairings
Ref [23]	7.588	90.000	3 ECM + 2 pairings
Ref [24]	2.352	2.900	1 ECM + 4 hash
Ref [25]	1.245	1.600	1 ECM + 3 hash
Proposed	**400**	**500**	**1 ECC + 2H + 4 AES**

Protocol	Energy	vs Proposed
Ref [22]	~2300 uJ	47x more
Ref [23]	~8900 uJ	182x more
Ref [24]	~280 uJ	5.7x more
Proposed	**48.8 uJ**	**baseline**

Energy per Authentication Cycle

Scyther Verification

Scyther is an automated security protocol verification tool, it performs exhaustive bounded verification.

The tool checks security claims including:

- **Alive: The claimed peer actually executed the protocol**
- **Weakagree: The peer agrees on the protocol execution**
- **Niagree: Non-injective agreement -- both parties agree on all data values**
- **Secret: The specified value remains unknown to the attacker**

Protocol file: uwc_protocol.spdl
Settings: max-runs = 5

Execution time: 0.19 seconds
Result: 32/32 claims verified -- zero attacks found

Hop	Initiator	Alive	Weakagree	Niagree	Secret	Responder	Alive	Weakagree	Niagree	Secret
1	UWS	Ok	Ok	Ok	Ok	SUB	Ok	Ok	Ok	Ok
2	SUB	Ok	Ok	Ok	Ok	BUOY	Ok	Ok	Ok	Ok
3	BUOY	Ok	Ok	Ok	Ok	SAT	Ok	Ok	Ok	Ok
4	SAT	Ok	Ok	Ok	Ok	BS	Ok	Ok	Ok	Ok

Security Analysis

Attack Defended	Defense Mechanism	Verified By
Replay	Fresh nonce + timestamp (delta t < 5s) per hop	Simulation + Scyther
Man-in-the-Middle	ECDH key agreement -- only recipient with sk can decrypt	Scyther (Niagree)
Impersonation	Node ID = SHA256(public key) -- computationally unforgeable	BAN Logic
Eavesdropping	AES-GCM ciphertext + ECDH ephemeral key exchange	Scyther (Secret)
Node Compromise	Ephemeral session keys + revocation list	BAN Logic
Delay Injection	Z-score anomaly detection on delay stream	Simulation (new)
Packet Injection	AES-GCM 128-bit authentication tag	Cryptographic proof

Z-score–based anomaly detection identifies delay-injection attacks as statistical outliers, achieving ~99% detection accuracy with low false positives (<1.2%).

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THANK YOU